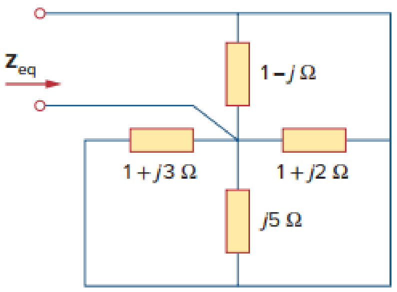
Problem

Find **Z**eq in the circuit of Fig. 9.68.

Figure 9.68:



Step-by-step solution

Step 1 of 1

All of the Impedances are in parallel,

$$\frac{1}{Z_{eq}} = \frac{1}{1-j} + \frac{1}{1+j2} + \frac{1}{j5} + \frac{1}{1+j3}$$

$$\frac{1}{Z_{eq}} = (0.5+j0.5) + (0.2-j0.4) + (-j0.2) + (0.1-j0.3)$$

$$= 0.8 - j0.4$$

$$Z_{eq} = \frac{1}{0.8-j0.4} = 1 + j0.5\,\Omega \ .$$